Chapter 16: Composite Materials

ISSUES TO ADDRESS...

- What are composites
- What are the classes and types of composites?
- What are the advantages of using composite materials?
- How do we predict the stiffness and strength of the various types of composites?

Composite

- A composite is a material made from two or more different materials with different physical and chemical properties that, when combined, are stronger than those individual materials by themselves.

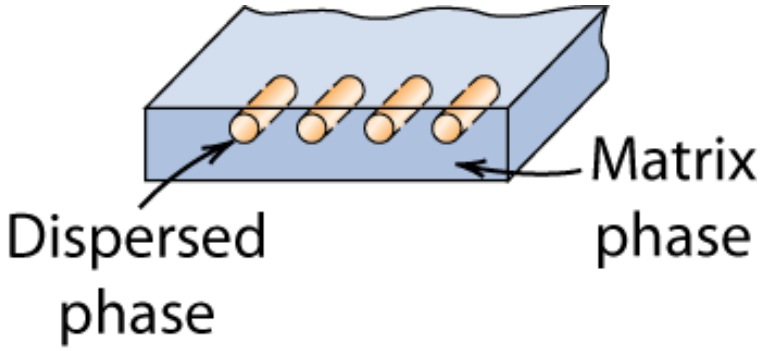
- Design goal: obtain a more desirable combination of properties (principle of combined action)
  - e.g., low density and high strength

Terminology/Classification

- Phase types:

  -- Matrix Material Phase - is continuous

  -- Dispersed material Phase- is discontinuous and surrounded by matrix



Adapted from Fig. 16.1(a), *Callister & Rethwisch 9e*.

The matrix is basically a homogeneous material in which a dispersed material of the composite is embedded. It is completely continuous.

The matrix provides a medium for binding and holding reinforcements together into a solid. It offers protection to the reinforcements from environmental damage, serves to transfer load, and provides finish, texture, color, durability and functionality.

The dispersed material or reinforcement is discontinuous and provides strength and stiffness

An example is the Fiber-Reinforced Polymer (FRP) composites, which are made from a polymer matrix that is reinforced with an engineered, man-made or natural fiber (like glass, carbon or aramid) or other reinforcing material.

The matrix protects the fibers from environmental and external damage and transfers the load between the fibers.

The fibers, in turn, provide strength and stiffness to reinforce the matrix—and help it resist cracks and fractures

Type of Composites Matrix Material

Ceramic matrix - Ceramic matrix composites (CMCs) are a subgroup of composite materials with ceramic matrix.

An example is the ceramic fibers embedded in a ceramic matrix, thus forming a ceramic fiber reinforced ceramic (CFRC) material.

CMC materials were designed to overcome the major disadvantages such as low fracture toughness, brittleness, and limited thermal shock resistance, faced by the traditional technical ceramics.

Metal matrix - Metal matrix composites (MMCs) are composite materials that contain at least two constituent parts – The metal matrix is reinforced with the other material to improve strength and wear.

Where three or more constituent parts are present, it is called a hybrid composite. In structural applications, the matrix is usually composed of a lighter metal such as magnesium, titanium, or aluminum.

In high temperature applications, cobalt and cobalt-nickel alloy matrices are common. Continuous carbon, silicon carbide, or ceramic fibers are some of the materials that can be embedded in a metallic matrix material.

MMCs are fire resistant, operate in a wide range of temperatures, do not absorb moisture, and possess better electrical and thermal conductivity. They have also found applications to be resistant to radiation damage. Most metals and alloys make good matrices for composite applications.

Polymer matrix - Polymer matrix composites (PMCs) can be divided into three sub-types, namely, thermoset, thermoplastic, and rubber.

PMC's consist of a polymer matrix combined with a fibrous reinforcing dispersed phase.

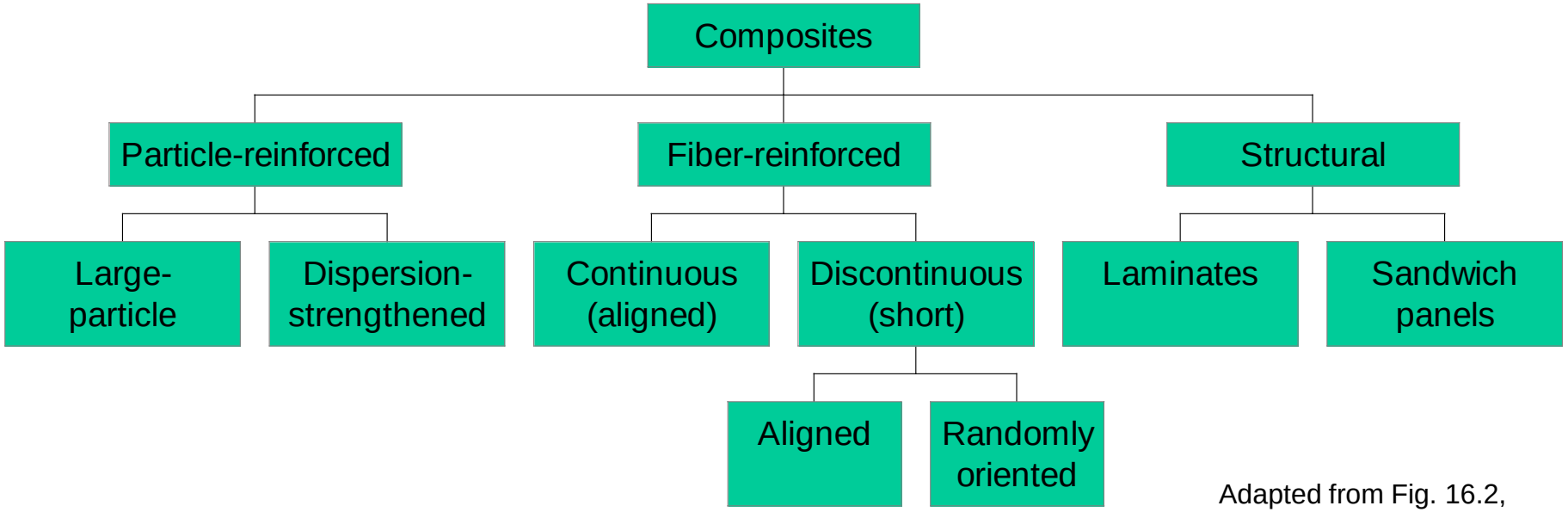
They are cheaper with easier fabrication methods. PMC's are less dense than metals or ceramics, can resist atmospheric and other forms of corrosion, and exhibit superior resistance to the conduction of electrical current.

Application of Composites Matrix Material

The following are common application areas of composite matrix materials:

- High performance Cookware
- Sealants and gaskets
- Heat shield systems (capable of handling high temperatures, thermal shock conditions and heavy vibration)
- Components for high-temperature gas turbines such as combustion chambers, stator vanes and turbine blades
- Brake disks and brake system components used in extreme thermal shock environments
- Components for burners, flame holders, and hot gas ducts

Classification of Composites



Adapted from Fig. 16.2, *Callister & Rethwisch 9e*.

Particulate-Reinforced Composites

Particulate-reinforced composites are composed of particulates embedded in a matrix material.

Particle reinforcing in composites is a less effective means of strengthening than fiber reinforcement.

Particulate reinforced composites achieve gains in stiffness primarily, but also can achieve increases in strength and toughness.

In all cases the improvements are less than would be achieved in a fiber reinforced composite.

Particulate reinforced composites find applications where high levels of wear resistance are required such as road surfaces.

The principal advantage of particle reinforced composites is their low cost and ease of production and forming.

Rule of Mixture

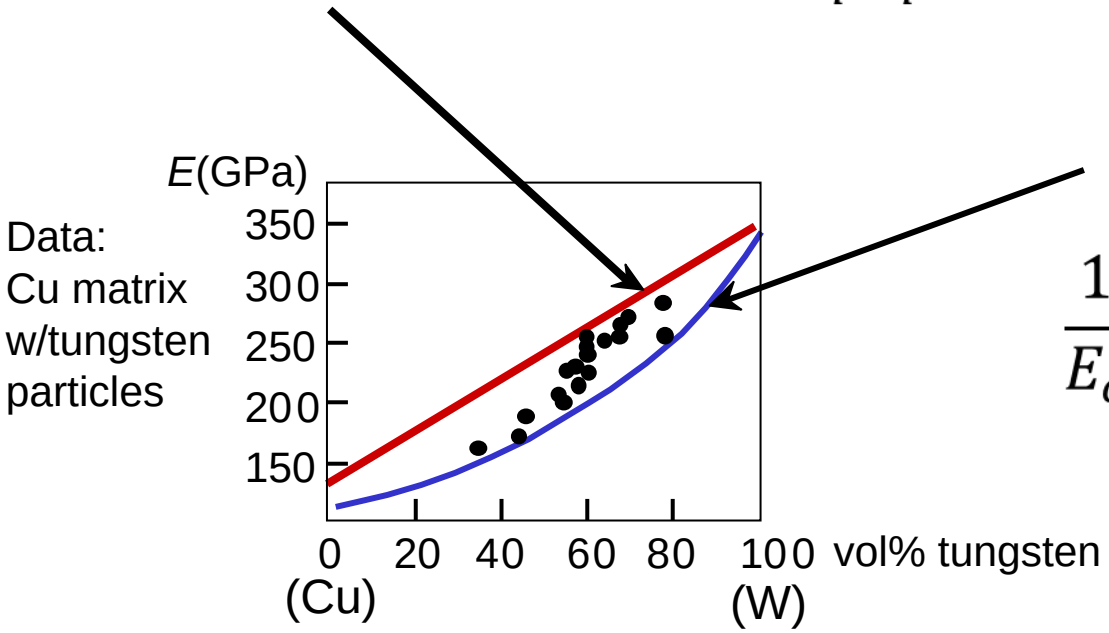
In material science, a general rule of mixtures is the weighted mean used to predict various properties of a composite material.

It provides a theoretical upper- and lower-bound on properties such as the
Elastic Modulus (E),
Mass density (ρ),
Ultimate Tensile Strength (UTS),
Thermal Conductivity (k) and
Electric Conductively (σ_e).

Rule of Mixture

- Elastic modulus, E_c, of composites:

UPPER LIMIT: $E_c = V_m E_m + V_p E_p$



LOWER LIMIT:

$$\frac{1}{E_c} = \frac{V_m}{E_m} + \frac{V_p}{E_p}$$

- Application to other properties:
  -- Electrical conductivity, σ_e: Replace E's in equations with σ_e's.
  -- Thermal conductivity, k: Replace E's in equations with k's.

Fiber-Reinforced Composites

Fiber is a natural or artificial substance that is significantly longer than it is wide with an average diameter measured in millionth (10^{-6}) of a meter.

Fiber-reinforced composites are composed of axial particulates embedded in a matrix material. The objective of fiber-reinforced composites it to obtain a material with high specific strength.

The strength is obtained by having the applied load transmitted from the matrix to the fibers.

Types of Fiber Arrangement

Aligned

The longitudinal tensile strength will be high whereas the transverse tensile strength can be much less than even the matrix tensile strength. It will depend on the properties of the fibers and the matrix, the interfacial bond between them, and the presence of voids.

- **Random**

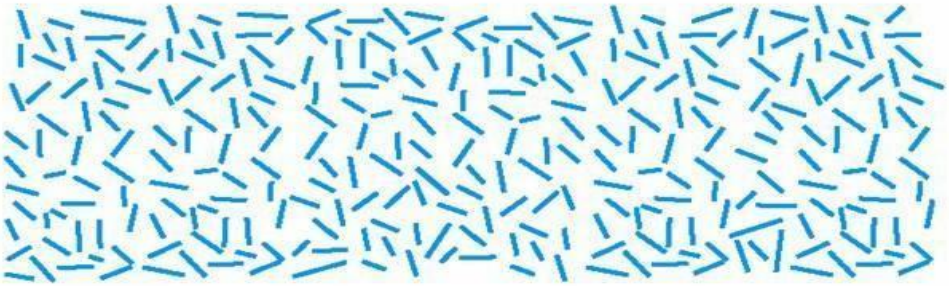
This is also called discrete, (or chopped) fibers. The strength will not be as high as with aligned fibers, however, the advantage is that the material will be istropic and cheaper.

- **Woven**

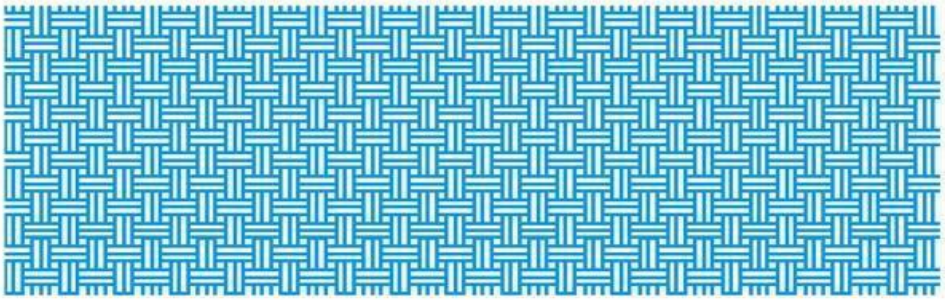
The fibers are woven into a fabric which is layered with the
- matrix material to make a laminated structure



Aligned Fibers



Random Fibers



Woven Fibers

Types of Fibers

Whiskers

very small diameter (~1 micron) single crystals

- strong because they are virtually flaw free
- expensive
- difficult to put in a matrix
- examples include graphite (C), SiN, Al_2O_3, SiC

- **Fibers**

small diameters (~10 microns)

- can be polycrystalline or amorphous

- **Wires**

large diameters (~25 microns)

- made from metals such as steel, Mo, W

- Fiber
  - Provide significant strength improvement to the composite

Example: An example is the Fiber-Reinforced Polymer (FRP) composites, which are made from a polymer matrix that is reinforced with an engineered, man-made or natural fiber (like glass, carbon or aramid) or other reinforcing material.

- Glass fibers
  - strength and stiffness
- Polymer matrix
  - holds fibers in place
  - protects fiber surfaces
  - transfers load to fibers

Classification: Structural

Particle-reinforced | Fiber-reinforced | **Structural**

A structural composite is a multi-layered and normally low-density composite.

They are used in applications requiring structural integrity, ordinarily high tensile, compressive, and torsional strengths and stiffnesses.

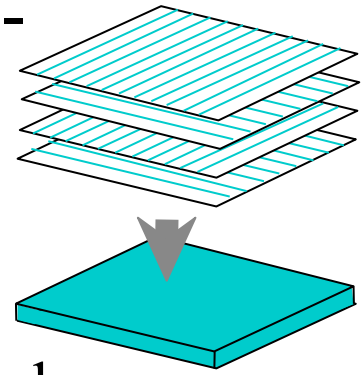
The properties of these composites depend not only on the properties of the constituent materials, but also on the geometrical design of the structural elements.

Laminar composites and sandwich panels are two of the most common structural composites

Classification: Structural

Particle-reinforced | Fiber-reinforced | **Structural**

A laminar composite is composed of two-dimensional sheets or panels bonded to one another.



Adapted from Fig. 16.16, *Callister & Rethwisch 8e*.

A Sandwich composite consists of two outer Sheets, that are separated by and adhesively bonded to a thicker core

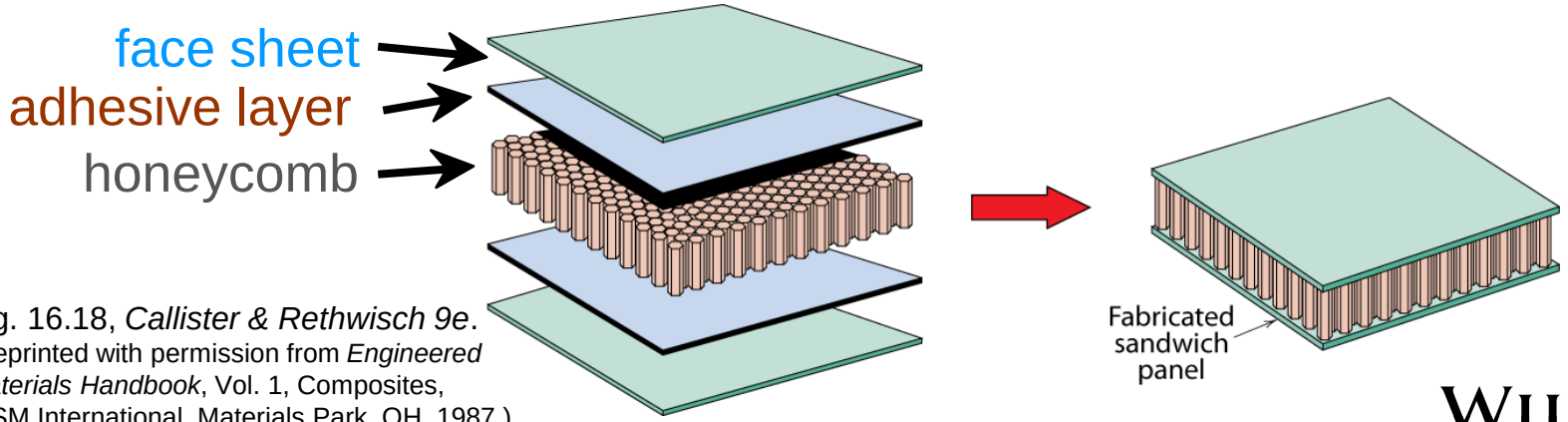


Fig. 16.18, *Callister & Rethwisch 9e*.
(Reprinted with permission from *Engineered Materials Handbook*, Vol. 1, Composites, ASM International, Materials Park, OH, 1987.)

Summary

- Composites types are designated by:
  -- the matrix material (CMC, MMC, PMC)
  -- the reinforcement (particles, fibers, structural)
- Composite property benefits:
  -- MMC: enhanced E, σ^*, creep performance
  -- CMC: enhanced K_{Ic}
  -- PMC: enhanced E/ρ, σ_y, TS/ρ
- Particulate-reinforced:
  -- Types: large-particle and dispersion-strengthened
  -- Properties are isotropic
- Fiber-reinforced:
  -- Types: continuous (aligned)
  discontinuous (aligned or random)
  -- Properties can be isotropic or anisotropic
- Structural:
  -- Laminates and sandwich panels

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: